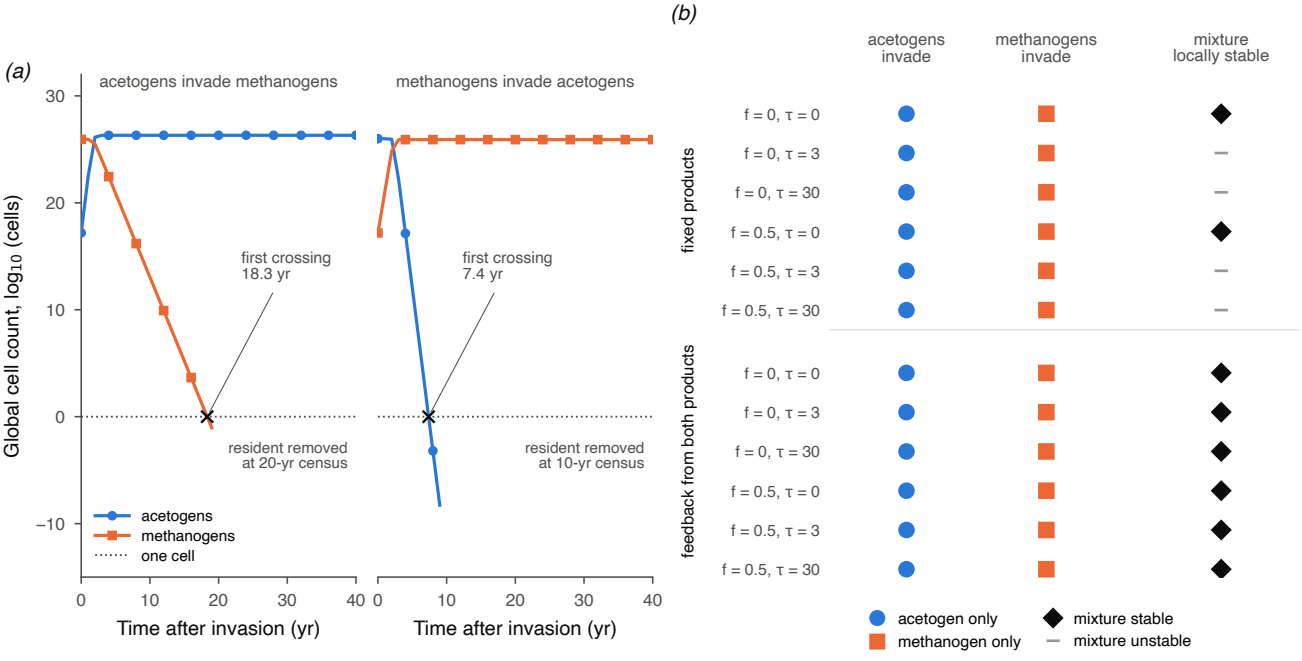
(a)
acetogens invade methanogens
methanogens invade acetogens
Global cell count, log10 (cells)
30
20
10
0
−10
first crossing
18.3 yr
first crossing
7.4 yr
resident removed
at 20-yr census
resident removed
at 10-yr census
acetogens
methanogens
one cell
0
10
20
30
40
Time after invasion (yr)
Time after invasion (yr)
(b)
acetogens
invade
methanogens
invade
mixture
locally stable
fixed products
f = 0, τ = 0
f = 0, τ = 3
f = 0, τ = 30
f = 0.5, τ = 0
f = 0.5, τ = 3
f = 0.5, τ = 30
feedback from both products
f = 0, τ = 0
f = 0, τ = 3
f = 0, τ = 30
f = 0.5, τ = 0
f = 0.5, τ = 3
f = 0.5, τ = 30
acetogen only
methanogen only
mixture stable
mixture unstable